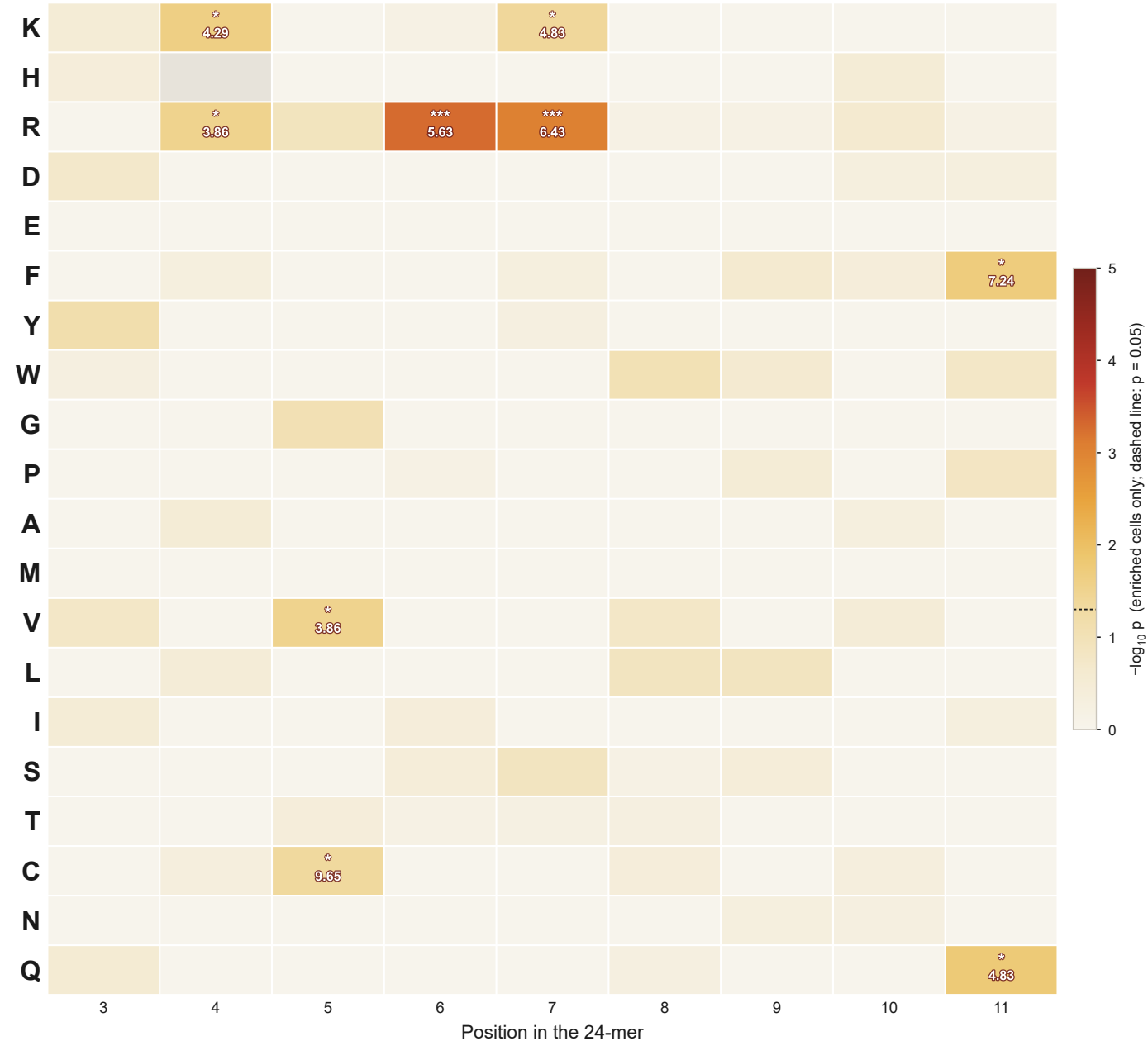
Position × residue enrichment, coloured by significance (enriched cells only) (positions 3–11)

same layout as the original Colab heatmap, but the colour scale is $-\log_{10}$ p (Fisher exact) and depleted cells are held at the floor



n = 20 P/G-D/E/T substrates vs n = 193 non-substrate controls carrying the same motif. Rows are ordered by chemical class — Basic K H R · Acidic D E · Aromatic F Y W · Aliphatic G P A M · Hydrophobic V L I · Polar S T C N Q.

Cell text is the fold change (substrates / controls) of every cell at p < 0.05, under the star bin of its uncorrected p: * p < 0.05, ** p < 0.01, *** p < 0.001. Grey = residue absent from both groups, no test possible.

No depleted cell reaches p < 0.05 anywhere in this dataset, so every coloured cell above is an enrichment. 9 of the 179 testable cells shown here reach p < 0.05 uncorrected, against 9 expected by chance alone — and none survives FDR correction, so no single cell here is more than a multiple-testing artefact.

This panel is the N-terminal window only — positions 3–11, out to the 11th amino acid of the 24-mer. The crop is a display crop: every cell's p is unchanged, and q is still BH-FDR across all 416 cells of the full 3–23 matrix rather than recomputed inside the window — recomputing it would be choosing the region after seeing the data.